Machine Learning-Based Classification of Ice Hockey Skating Tasks Using Kinematic Data

Oussama Jlassi[a*], Ethan W.C. Wilkie[a], Matthew Kelly[a], Philippe J. Renaud[a], David J. Pearsall[a], Shawn M. Robbins[b], Philippe C. Dixon[a*]

[a]*Department of Kinesiology and Physical Education, McGill University, Montreal, Canada*

[b]*Centre for Interdisciplinary Research in Rehabilitation, Lethbridge-Layton-MacKay Rehabilitation Centre, and the School of Physical and Occupational Therapy, McGill University, Montreal, Canada*

*Corresponding author: philippe.dixon@mcgill.ca

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Abstract

This study evaluates the ability of body segment kinematic data to identify skating tasks in ice hockey using machine learning models and compares the performance of models trained on different body segments. We employed XGBoost, Support Vector Machine, and Random Forest models to classify four primary ice-hockey skating tasks: forward skating start and strides, skating stop & go, and skating into a wrist shot. Trunk, pelvis, thigh, shank, and foot segment center of mass linear accelerations were derived from retro-reflective markers and used as inputs for feature engineering. The models were trained and evaluated using a 10-fold cross-validation stratified by participant. Overall, the machine learning models demonstrated strong performance, with mean accuracy scores ranging from 86.5% to 98.9%. The pelvis yielded the best overall performance, followed by the trunk and foot, whereas the thigh segment generally exhibited lower accuracies across models. These results indicate that prediction performance depends on the body segment kinematic data used as input. This study highlights the potential of body segment kinematic data for automated identification of ice hockey skating tasks, providing insights into sports analytics and player performance assessment.

Keywords: Ice Hockey, Skate, Event detection, Machine Learning

1. Introduction

In today's rapidly evolving sporting landscape, ice hockey exemplifies how advancements in equipment and smart technologies are revolutionizing sport. These advancements have ushered in an era where sports analytics substantially enhance coaching strategies through the effective utilization of data. Motion data, in particular, now provides comprehensive and actionable metrics such as player speed, distance covered, and external workload estimates[1–5], enabling coaches and sport scientists to quantitatively assess performance, monitor fatigue, and tailor training programs to reduce injury risk and enhance athletic output. Among available tools, marker-based motion capture remains the gold standard for obtaining precise, high-resolution kinematic data from hockey players, but its use is limited by the need to attach markers directly to the athlete. Wearable sensors face similar constraints, requiring placement on the body, and only offer indirect estimates of position from acceleration. More recently, video-based motion capture has emerged as a promising alternative, leveraging computer vision to extract positional data directly and non-invasively, in ways that parallel marker-based systems. Industry partnerships are already accelerating the integration of video-based capture into professional hockey arenas, underscoring its potential for large-scale, in-game application.

Despite progress in data acquisition and analysis, a critical gap remains: the automation of task identification. Currently, segmenting motion data into discrete tasks (e.g., skating starts, turns, stops) is labour-intensive. Automating this process through machine learning would not only expedite data workflows but also enhance the interpretability of composite metrics like player load by anchoring them in task-specific context. This has motivated a growing body of research into machine learning-based movement classification in ice hockey.

Efforts to utilize artificial intelligence algorithms for movement analysis in ice hockey began as early as 2009[6]. Initial attempts used computer vision techniques such as Hue-Saturation-Value (HSV) colour filtering and Histograms of Oriented Gradients (HOG) features to detect players and classify actions from video footage[6]. HSV models distinguish colour differences under varying lighting[7], while HOG features capture edge orientations that reflect body posture and movement direction[8]. Despite their conceptual appeal, these methods achieved limited accuracy (50–75%) in identifying skating direction (i.e., forward and backward)[6], largely due to variability in player appearance and camera angles. Later studies incorporated body segment kinematics and deep learning architectures to enhance action recognition, expanding task identification to turns, crossovers, and stopping[9,10]. These models demonstrated improved classification performance but still faced issues of limited sample sizes, contributing to poor generalizability from controlled settings to real game conditions. Despite these limitations, the use of body segment kinematics remains a promising approach for automating the identification of discrete skating tasks. Prior ice hockey research has delineated key biomechanical signatures, such as hip extension, trunk lean, and knee extension velocity for acceleration; or ankle dorsiflexion, pelvic tilt, and trunk rotation for directional changes that are associated with skating starts, strides, and turns[11–17]. These insights support the development of machine learning models that classify tasks based on segmental motion data.

Although prior studies have demonstrated promising accuracy in classifying hockey tasks using foot segment kinematics (≥0.7 F1 scores)[9], the optimal combination of body segments for effective classification remains uncertain. Advancing this line of work requires not only maximizing model performance but also ensuring that selected segments are pragmatically suited for use in real-world, on-ice settings, favouring configurations that minimize occlusion and are compatible with game conditions.

Insights from related sports with analogous movement patterns, such as cross-country skiing, highlight the critical role of segment selection and marker placement in optimizing task classification accuracy[18–20]. These findings underscore the need for thoughtful segment selection and provide a transferable framework for refining classification models in ice hockey. Nevertheless, persistent challenges, such as limited participant diversity and model generalizability across varied skating styles continue to constrain the robustness and applicability of current approaches.

Thus, the primary objective of this study was to determine the accuracy of machine learning models (eXtreme Gradient Boosting (XGBoost), Support Vector Machines (SVM), and Random Forest (RF)) in classifying four different hockey tasks using segment center of mass (COM) acceleration computed from retro-reflective marker data. By incorporating a large and diverse participant cohort, this study also aimed to address limitations of previous work by introducing greater inter-individual variability, thereby enhancing the generalizability and robustness of model training. The secondary objective was to assess the relative performance of models based on individual segment COM data. We hypothesized that (1) models would achieve accuracy scores greater than 80% for skating task identification and (2) lower extremity segments (foot, shank, thigh) would outperform axial skeleton segments (trunk and pelvis).

2. Materials and methods

2.1. Participants and dataset description

Data from previous studies[17,21–25], captured using multiple infrared cameras on an arena ice surface (Vicon Motion Systems Ltd., Oxford, UK) at 240 Hz, were extracted from our laboratory database. All participants signed written informed consent documents allowing secondary analyses of the data. All studies were approved by the local ethics committee. The dataset includes 103 participants (72 males, 31 females) with varying levels of ice hockey experience to enhance

generalizability. Participants were classified as high-calibre (60 males and 31 females) if they played in either major junior (Canadian Hockey League) and/or the Canadian Interuniversity Sport hockey league. Low-calibre (12 males) participants played in any level lower than the aforementioned leagues.

Across all datasets, the recorded events include "Stop & Go," "Stride," "Start," and "Wrist shot." These tasks were selected because they represent biomechanically distinct movement patterns that are fundamental to ice hockey performance, occur frequently during gameplay, and are relevant for applications in player monitoring and skill classification. Not all participants performed all four tasks; rather, each of the seven contributing datasets was designed around specific tasks. For instance, the "Stop & Go" task was recorded in the Hallihan (2018) dataset, while "Stride" tasks were included in both the Budarick et al. (2018) and an unpublished dataset from Renaud et al. (2017). "Start" tasks were captured in three separate datasets: Shell et al. (2017), Renaud et al. (2017), and McPhee (2018). Finally, "Wrist shot" data were obtained from Robbins et al. (2020). This task-specific structure means that individual participants typically completed only one or two of the skating tasks, depending on the focus of the original study from which their data were drawn.

Marker placement varied slightly across datasets, reflecting differences in study objectives and capture setups. Most datasets used between 24 and 81 reflective markers placed on anatomical landmarks to define trunk and lower limb segments. Hallihan (2018), Budarick et al. (2018), and Shell et al. (2017) employed 51, 67, and 81 markers respectively, enabling detailed segmental tracking. Renaud et al. (2017) and its unpublished extension used a simplified configuration with 24 markers to capture essential lower body movements over a smaller capture volume. McPhee (2018) applied 48 markers, while Robbins et al. (2020) used 79 markers, incorporating both lower and upper body points to capture wrist shot mechanics.

2.2. *Data processing and biomechanical modelling*

First, all marker trajectories were reviewed and gap-filled (spline, Woltring, rigid-body), if necessary, in Vicon Software. Next, c3d files were imported into Visual3D (HAS-motion, Kingston, Canada) where marker data were filtered (cut-off frequencies ranging from 6 to 25 Hz) and used to compute segment COM positions from Dempster's regression tables[26], which define the relative location of the COM within each segment using normalized segment lengths and assumed mass distribution ratios. All marker sets enabled computation of the foot, shank, thigh, and pelvis segment COM. The trunk segment was included in all datasets except: the Start task from Renaud et al., 2017 and the Stride task from unpublished data in Renaud et al. (2017)). Although marker placement and filtering cut-off frequencies varied slightly across contributing datasets, all were within accepted biomechanical ranges and applied consistently within each dataset. To minimize the influence of these differences, we restricted our analysis to COM accelerations, which are less sensitive to marker placement variability than joint center definitions. This approach ensures comparability across datasets while acknowledging that some residual variability in pelvis COM estimates may remain, particularly in high-flexion movements such as skating. Then, segment COM acceleration data were imported into Matlab (The Mathworks Inc., Natick, USA) where custom scripts based on the biomechZoo toolbox[27] normalized each trial so that the initiating foot was designated as side 1 (S1) and the following foot as side 2 (S2). This convention was adopted to facilitate within-trial consistency across participants, since not all datasets consistently labelled left and right sides.

It is important to note that, while necessary for harmonization across heterogeneous datasets, the S1/S2 assignment effectively provides the model with knowledge of which foot initiates the movement. Therefore, model results reflect performance under conditions where initiation side is known, rather than the fully autonomous performance that would be expected in real-world applications. We then computed linear acceleration time series data in the global X, Y, and Z

directions by double differentiating the displacement of the COM for both trunk and pelvis segments, and bilaterally for thigh, shank, and foot segments. The global coordinate system was consistently defined across all datasets, with the X-axis oriented laterally (left–right across the rink), the Y-axis aligned with the longitudinal direction of the rink (forward–backward skating direction), and the Z-axis oriented vertically (upward). Each trial was exported as a separate comma-separated values (CSV) file.

2.3. Feature engineering

Feature engineering was performed prior to model development using Python (v. 3.10, Python Software Foundation) packages Pandas[28], Numpy[29] and Scikit-learn[30]. The rationale behind the feature engineering strategy was to capture biomechanically relevant patterns of movement across segments using both directional and non-directional acceleration data. Specifically, for each trial, raw COM acceleration signals in the X, Y, and Z directions were used to derive statistical (e.g., mean, standard deviation), time-domain (e.g., peak, range), and frequency-domain (e.g., spectral entropy, dominant frequency) features that could represent variability, intensity, and periodicity in movement. Additionally, a non-directional feature root-mean-square (RMS) acceleration was computed from the resultant acceleration vector to summarize overall movement magnitude, regardless of direction. These features were extracted from eight anatomical segments (trunk, pelvis, bilateral thighs, shanks, and feet), resulting in a total of 224 features (8 segments × 3 directions × 9 features + 8 segments × 1 RMS feature) as detailed in Table 1.

2.4. Feature reduction

We employed filter-based feature reduction techniques to improve model generalization, reduce redundancy, and streamline training. Features with high pairwise correlation (threshold ≥ 0.8) were considered redundant; within each correlated group, only the first

feature based on its order of appearance in the dataset was retained. Features showing no variation or extremely low variability (coefficient of variation < 0.1) across samples were also discarded, as they provide little discriminative value and may introduce noise. These filter methods are fast, scalable, and independent of the learning algorithm, making them ideal for preprocessing high-dimensional biomechanical data.[31] Each remaining feature's meaning is presented in Table 3.

2.5. Machine learning models

We developed 18 feature-based machine learning models using features extracted from all segments as well as from individual trunk, pelvis, thigh, shank, or foot segments. For each, 3 model types were implemented: XGBoost[32], SVM, and RF[33]. The dataset was partitioned using a 10-foldcross-validation strategy (StratifiedGroupKFold from Scikit-learn[30]). This ensured that all data from a single participant were assigned exclusively to either the training or test folds, preventing data leakage across splits. Features were scaled from 0 to 1 using a MinMaxScaler within the training folds to satisfy the requirements of the SVM, which is sensitive to feature magnitudes due to its distance-based computations. While RF and XGBoost are tree-based models and generally robust to unscaled features, scaling was applied uniformly to maintain consistency across models. Model training and evaluation were performed within each fold, and performance was summarized as the mean accuracy ± standard deviation across the five cross-validation splits. Feature importance was calculated for the top performing model in each category using features derived from all segments to demonstrate the relevance of each feature in differentiating tasks[32,33]. XGBoost calculates feature importance using metrics such as Gain, Cover, and Frequency. In this

study, Gain was used, which measures the improvement in accuracy brought by a feature to the branches it is on. RF, on the other hand, uses Gini importance (also known as Mean Decrease in Impurity) to calculate feature importance, which is the default method in most implementations. In this study, we used the default approach based on Gini impurity. This method measures the total decrease in node impurity (Gini impurity) brought by a feature, weighted by the probability of reaching that node, and summed across all trees in the forest. It provides an efficient way to rank features by their overall contribution to improving classification purity during training. SVM, calculate feature importance based on the coefficients of the hyperplane that separates the classes. These coefficients indicate the weight or importance of each feature in the decision boundary, with larger absolute values suggesting higher importance.

3. Results

The accuracy scores of the XGBoost, SVM, and RF models across different segment configurations are summarized in Table 2. Overall, RF and XGBoost consistently outperformed SVM across most configurations, with RF achieving the highest accuracy in several cases. Notably, the pelvis and trunk segment configurations yielded the strongest results, with RF reaching 98.9% and 98.4% accuracy, respectively. XGBoost also performed very well on the pelvis (98.3%) and trunk (97.2%), while SVM achieved its best performance on the pelvis (98.4%). For the all-segments configuration, RF and XGBoost maintained high accuracies of 97.2% and 96.7%, respectively, outperforming SVM (94.9%). Although SVM lagged behind on the thigh (86.5%) and trunk (87.7%), it remained competitive on the shank (96.3%) and foot (96.5%), showing that performance varied depending on the segment. The top five ranked features contributing to each model's performance are illustrated in Figure 1, highlighting the segments and axes most critical for distinguishing skating tasks.

4. Discussion

This study demonstrates the feasibility of using body segment COM acceleration data with machine learning to accurately classify discrete ice hockey skating tasks. High performance across all models, particularly XGBoost and RF models, which achieved perfect accuracy using all segment data (trunk, pelvis, thighs, shanks, feet), underscores the potential for segmental motion data to support automated task recognition in on-ice settings.

Model performances were excellent, surpassing our hypothesized threshold of 80% accuracy, thus supporting our first hypothesis. Perfect accuracy achieved by the ensemble-based models is promising, but should be interpreted with caution. Without repeated evaluations on independent datasets, the robustness of these results cannot be guaranteed. While the large and diverse participant sample likely contributed to improved generalizability and helped address limitations noted in prior work[9], it is unlikely the sole factor. The integration of comprehensive feature engineering and the use of multi-segment input data likely also played a significant role in driving model performance. Future work should assess performance in real-time, continuous gameplay data streams. Although both ensemble models performed equally well in our controlled dataset, their relative strengths may become more apparent under in-game conditions where movement patterns are less discrete, transitions are fluid, and external conditions are more variable. XGBoost, which builds decision trees sequentially using boosting[32], may be better suited to capture the complexity of in-game movement transitions, such as distinguishing a pivot into a shot from a standard stop. RF, by contrast, builds trees in parallel[33] and is typically faster to train and easier to interpret, which could be advantageous in lower-latency applications like on-bench monitoring or for quick post-game feedback. The choice between these approaches will ultimately depend on the deployment constraints and complexity of the input data, and warrants further investigation.

While our second hypothesis indicated that lower limb segments would outperform axial segments, the pelvis emerged as the strongest individual contributor to model performance. Interestingly, despite its high accuracy, pelvis-derived features were not among the most influential in the importance rankings (Figure 1). This discrepancy suggests that pelvis kinematics may provide globally informative, though less isolated, cues that serve as a proxy for whole-body motion; however, interpretation is complicated by methodological challenges in defining pelvis kinematics with retro-reflective markers in high-flexion sports such as ice hockey. Soft-tissue artifact, difficulties in consistently tracking pelvic landmarks, and potential occlusion in crouched or rotated positions can all influence the reliability of pelvis-based signals.

Moreover, because our dataset aggregated multiple studies with slightly different marker placements and filtering parameters, some residual variability in pelvis COM estimates is possible despite our decision to focus on accelerations rather than joint centers. As such, while the pelvis may represent a robust global marker of movement, its role should be interpreted with caution, and future work should validate whether pelvis-driven accuracy reflects genuine biomechanical signal or methodological artifact. In alignment with Vahdatpour et al. (2011) and Panebianco et al. (2018), the foot and shank segments consistently ranked among the top contributors to model accuracy (Figure 1), highlighting their utility in capturing fine-grained, task-relevant kinematic features. This also aligns with prior findings in both ice hockey and related sports such as cross-country skiing, where distal limb wearable sensors are favored for capturing task-specific dynamics[9,10,18–20]. Their strong performance may stem from the transition from unimodal to bimodal skating patterns during forward accelerations and decelerations[22], which manifests most prominently in the distal lower limbs.

Interestingly, while thigh-derived features were listed among the most important across both XGBoost and RF models (Figure 1), the segment as a whole yielded lower classification accuracy

compared to the foot and shank. This may reflect a discrepancy between feature salience and segment-level classification capacity. A small number of highly informative thigh features (e.g., mean Y-plane accelerations) may play a critical role in certain decision boundaries, while the segment's overall motion profile may be less distinct across skating tasks. In contrast, the foot and shank exhibit richer dynamic variability during task transitions, offering a more discriminative input signal despite individual features being ranked slightly lower.

Although this study relied on retro-reflective marker data, the segment-specific findings provide actionable guidance for both research and the applied development of wearable and video-based motion capture systems. When considering practical deployment whether in marker-based research, inertial measurement unit (IMU) monitoring, or video-based motion capture, both the foot and pelvis segments present distinct advantages and limitations. The foot provides rich task-specific information, but is mechanically vulnerable in real-world gameplay and more prone to occlusion. In video-based capture, rapid foot movements may blur depending on camera frame rate, and sight lines may be blocked by sticks or opposing players. Similarly, foot-mounted IMUs can be displaced or damaged by slashes, blocked shots, or high-speed impacts. While these issues are negligible in controlled laboratory environments, they pose significant challenges for field-ready systems. By contrast, the pelvis represents a more secure and non-intrusive location. IMUs positioned at the sacrum remain stable during gameplay and are already commonly used in locomotion studies[34,35]. Likewise, for video-based capture, the pelvis is less susceptible to occlusion and motion blur than the distal limbs, offering a more reliable target for computer vision approaches. Nonetheless, the centrality of this segment may limit its sensitivity to the fine-grained kinematic differences that distinguish discrete skating tasks. Future work should therefore evaluate whether a single, strategically placed sensor, particularly at the pelvis, can provide sufficiently accurate classification for practical use. For video-based methods, multi-view capture setups and algorithms that combine

the stability of axial landmarks with the task specificity of distal segments may prove most effective in advancing scalable, in-game applications.

Despite these promising findings, key limitations remain. Our reliance on retro-reflective marker data restricts the direct translation of these models to in-game environments. It is important to emphasize that this work was designed as a proof of concept rather than a final application. Future studies should validate performance using signals collected from IMUs or video-based motion capture during continuous gameplay, where task boundaries are less distinct and transitions occur fluidly. Additionally, while our dataset was relatively large and diverse in skating tasks, it was skewed toward high-calibre and male participants. Prior studies have highlighted meaningful kinematic differences based on skill level and sex[22–24], suggesting that model performance should be further evaluated in youth, recreational, and female athlete populations to enhance generalizability and applicability. Moreover, we note that COM accelerations in this study were defined relative to a global rink-based coordinate system. While this facilitates comparison across trials and participants, it may not fully capture the directions of movement experienced during real gameplay, where athletes rarely move perfectly along the global axes. Defining accelerations relative to a local reference frame, such as the pelvis or trunk, could provide more ecologically valid insight into movement patterns and may be a useful approach in future work, particularly for models intended to operate in continuous, in-game scenarios.

An interesting note is that variability in marker placement and filtering across datasets may have inadvertently contributed to improved model generalizability. While we aimed to standardize preprocessing and focus on COM accelerations to mitigate methodological inconsistencies, the heterogeneity in data collection protocols could have provided additional variation that helped the models learn more robust patterns. This highlights both a potential benefit and a limitation of

aggregating multiple datasets and should be considered when interpreting the results.

Additionally, it is possible that certain skating tasks were inherently easier to classify than others. While our overall accuracy scores were high, misclassifications were not evenly distributed across tasks. Future work could perform a detailed task-level error analysis, such as examining confusion matrices or task-pair misclassification rates, to identify which movements are most challenging for the models. This would provide more nuanced insight into model strengths and limitations and inform the design of task-specific features or sensor placements.

Finally, while we adopted an S1/S2 convention to harmonize datasets with inconsistent labelling of left and right sides, we acknowledge that this is not directly transferable to autonomous, real-world applications, where the initiating foot is not known a priori. For practical deployment, models should instead operate on fixed left/right definitions.

Taken together, these findings advance the integration of machine learning into ice hockey performance analysis by identifying viable segment configurations and highlighting the potential for minimally intrusive sensor systems. As technologies continue to evolve, combining biomechanics and data science will be critical in developing scalable, field-ready tools for skill recognition and workload monitoring in ice hockey.

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Disclosure of interest

The authors declare no competing interests in this study

Author Contributions

Oussama Jlassi refined the machine learning models and drafted the final manuscript, Ethan W.C. Wilkie Interpreted the results and drafted the final manuscript, Matthew Kelly conducted the data inspection and model development as well as the drafting of the manuscript's first version. Philippe J. Renaud was involved in the conceptualization of the study, methodology and software. Shawn M. Robbins and David J. Pearsall were involved during the conceptualization, methodology, software, resources, writing-review and editing, supervision and funding acquisition. Philippe C. Dixon was involved in the methodology, software, writing-review and editing, and supervision. All authors reviewed and approved the final version of the manuscript.

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Tables and Figures:

Table 1: Features calculated for each segment and their respective categories

Feature	Statistical	Time Domain	Frequency
Mean	✓		
Minimum	✓		
Maximum	✓		
Standard Deviation	✓		
Skewness	✓		
Kurtosis	✓		
Root Mean Square		✓	
Resultant Root Mean Square*		✓	
Entropy		✓	
Mean Power Frequency			✓

*indicates the non-directional feature

Table 2: Machine learning model performance based on input segment configuration

Data	Model	Mean Accuracy (std)(%)
All segments	XGBoost	96.7 (0.05)
	SVM	94.9 (0.08)
	RF	97.2 (0.08)
Thigh	XGBoost	93.1 (0.02)
	SVM	86.5 (0.01)
	RF	93.3 (0.1)
Trunk	XGBoost	97.2 (0.06)
	SVM	87.7 (0.10)
	RF	98.4 (0.03)
Shank	XGBoost	96.1 (0.05)
	SVM	96.3 (0.04)
	RF	96.4 (0.05)

Foot	XGBoost	97.4 (0.03)
	SVM	96.5 (0.04)
	RF	97.1 (0.05)
Pelvis	XGBoost	98.3 (0.05)
	SVM	98.4 (0.05)
	RF	98.9 (0.05)

498 Abbreviations: eXtreme Gradient Boosting (XGBoost), Support Vector Machines (SVM),
499 and Random Forest (RF), standard deviation (std).

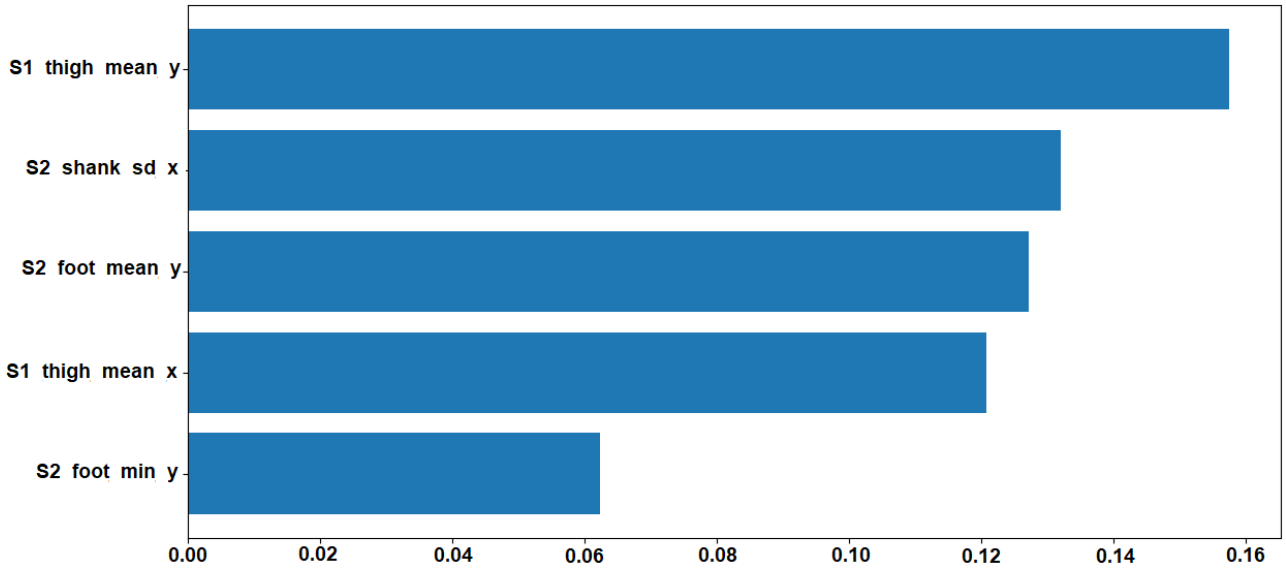
500 Table 3: Description of Features

501

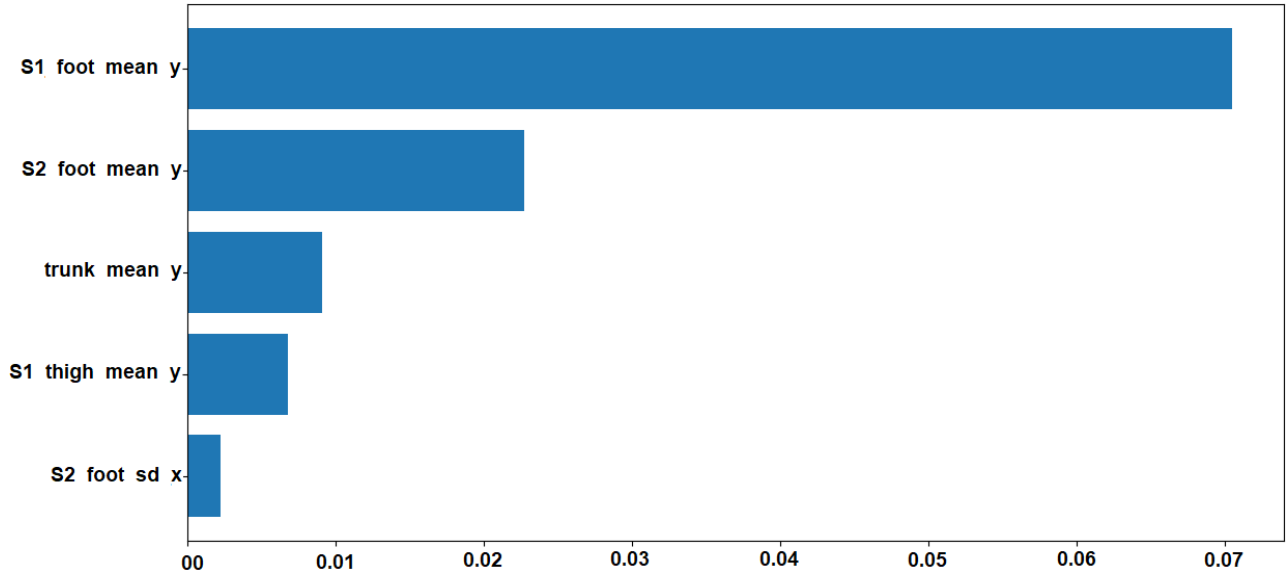
Feature	**Meaning**
S1 thigh mean y	Y component's mean acceleration of the thigh segment on side 1
S2 foot sd x	X component's acceleration's standard deviation of the foot segment on side 2
S2 foot mean y	Y component's mean acceleration of the foot segment on side 2
S1 foot sd y	Y component's acceleration's standard deviation of the foot segment on side 1
S1 foot mean y	Y component's mean acceleration of the foot segment on side 1
trunk mean y	Y component's mean acceleration of the trunk segment
S2 shank sd x	X component's acceleration's standard deviation of the shank segment on side 2
S1 thigh mean x	X component's mean acceleration of the thigh segment on side 1
S2 foot min y	Y component's minimum acceleration of the foot segment on side 2

502 The X and Y axes represent the horizontal and vertical directions within the 2-dimensional

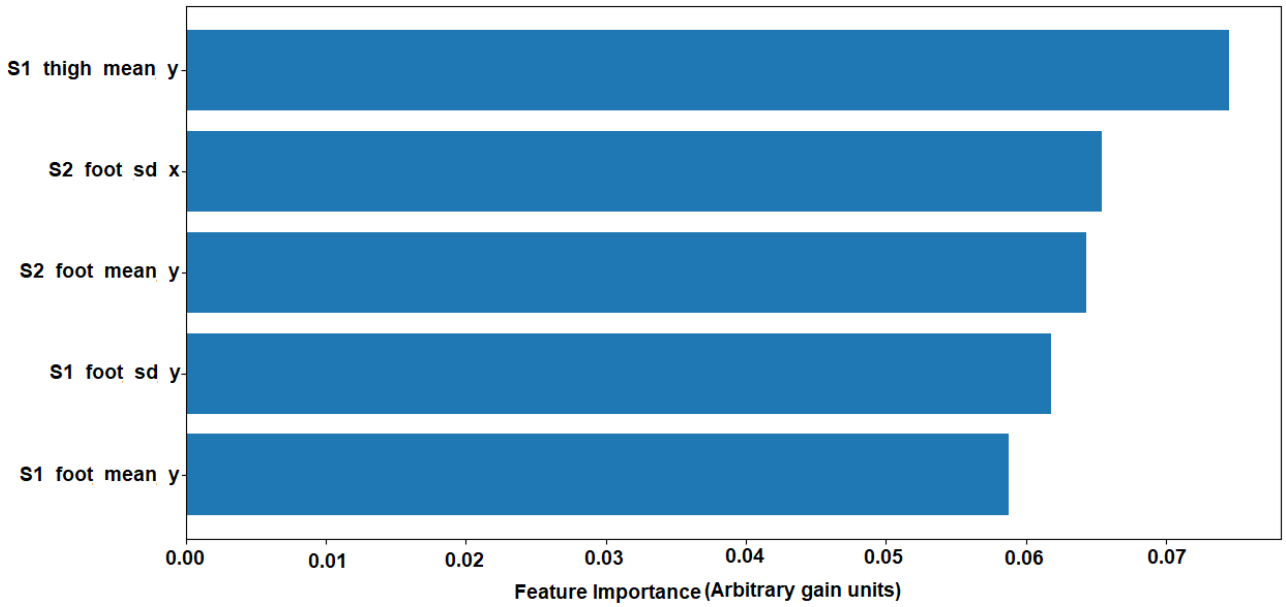
plane of the skating rink. Abbreviations: Side 1 (S1), side 2 (S2), standard deviation (sd), minimum (min), and maximum (max).



(a)



(b)



(c)

Figure 1

Figure captions:

Figure 1: Relative importance of the top 5 features for the (a) eXtreme Gradient Boosting (XGBoost), (b) Random Forest (RF), and (c) Support Vector Machines (SVM) model based on input features from all segments. The X and Y axes represent the horizontal and vertical directions within the 2-dimensional plane of the skating rink. Abbreviations: Side 1 (S1), side 2 (S2), standard deviation (sd), minimum (min), and maximum (max).